USCG CAPTAIN'S LICENSE · OUPV NEAR COASTAL

Deck General & Safety

Visual study guide — diagrams, memory hooks & exam traps



Firefighting · Lifesaving & survival · Man overboard · Boat handling · Anchoring
Marlinespike · Towing · Weather · Radio · Pollution · Engines & fueling · Six-pack rules
Built from 265 MLS Deck General cards + 48 official USCG Q170 sample questions
Prepared for Jesse Orebaugh · October 2026

How this exam module works

Deck General & Safety (DGS) is the biggest, broadest module on the OUPV exam: a little of everything a six-pack captain needs to keep people alive and the boat off the bottom. The passing mark is **70%** (Rules of the Road is the only 90% module). The questions are mostly recall, so this guide is built around the exact facts the question bank keys, grouped by topic.

- **Read the last words of the stem first.** "FIRST action," "NOT," "EXCEPT" and "LEAST" flip the answer. Circle them mentally before you look at the choices.
- **Pick the bank's answer, not the debate.** A handful of keys are dated or arguable. They're collected on the **Bank keys** page so you can memorize the expected answer and move on.
- **"All of the above" is often right in DGS** when every choice is a reasonable safety practice (anchor watch duties, liferaft equipment, ways to detect dragging).
- **Think like a pilot.** Many answers follow cockpit logic: alarm before troubleshooting, land into the wind (dock into the current), checklists before emergencies.

The module at a glance

Topic	MLS cards	What gets tested most	Page
Firefighting	42	Fire classes, tetrahedron, agents, CO2 hazards, extinguisher sizes	3
Six-pack regulations	5	Extinguishers by length, placards, checkoff list, PFD rules	5
Lifesaving equipment	30	PFD types, immersion suits, EPIRB, liferaft deployment	6
Survival & man overboard	18	Water rationing, hypothermia, Williamson vs racetrack	8
Boat handling	22	Prop walk, bank effect, squat, pivot point, docking, heavy weather	10
Anchoring & towing	21	Scope math, holding, dragging, catenary, bridles	12
Marlinespike	30	Bowline, sheet bend, splices, line types, deck terms	13
Weather	20	Fronts, veer/back, sea breeze, fog, Beaufort	14
Radio	19	MAYDAY / PAN-PAN / SÉCURITÉ, Ch 16 and 13, bridge-to-bridge	15
Pollution	18	Reporting, MARPOL garbage zones, MSD types, spill cleanup	16
Engines, fueling & watch	29	Gasoline vapor, blower, diesel ignition, lookout	17

MLS card counts by topic (11 more are untagged). The 48 official Q170 samples are spread across every topic.

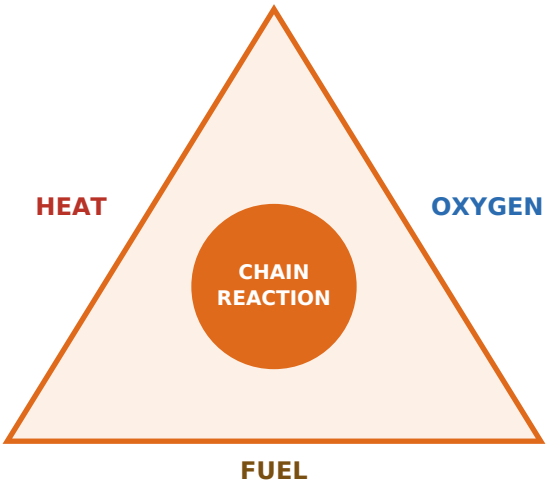
TIP **How to use this guide with the app**

Read one topic here, then in CA License Study filter **Deck Gen** + that topic in Quiz mode. Finish with **USCG Samples** (Q170) for the official wording. Your 17 missed MLS questions are on page 18.

Firefighting I: how fire works

The fire tetrahedron

The old **fire triangle** is heat, fuel and oxygen. The **tetrahedron** adds the fourth element: the self-sustaining **chemical chain reaction**. Every extinguishing method removes one of the four.



- **Remove HEAT → cool**
  Water (best), water fog/spray; cool adjacent bulkheads
- **Remove OXYGEN → smother**
  CO2, foam, fire blanket; secure ventilation
- **Remove FUEL → starve**
  Shut fuel valve; good housekeeping
- **Break the CHAIN → inhibit**
  Dry chemical (fastest knock-down), Halon-type

EXAM TRAP **Things that are NOT a side of the triangle or tetrahedron**

Electricity (it's an ignition source, not a side) and **pressure**. If a question asks what else is needed besides heat, fuel and oxygen, the answer is **chain reaction**.

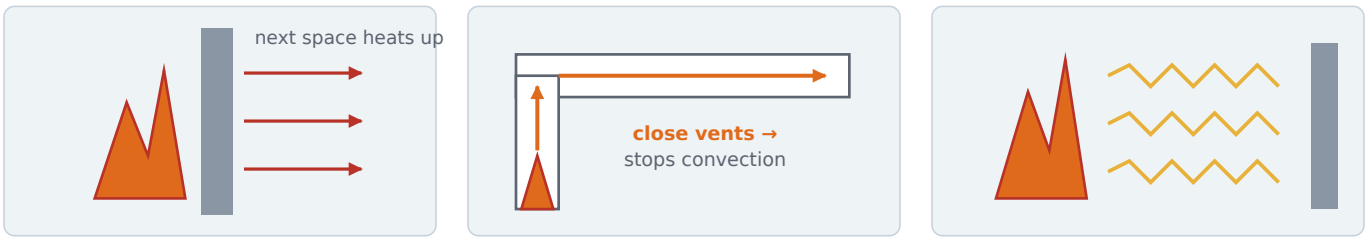
Fire classes

Class	Burning	Best agents	Hook
A	Ordinary combustibles: wood, paper, cloth, linen, trash	Water (straight stream breaks it up)	**A** = leaves **A**sh
B	Flammable liquids: gasoline, diesel, oil, grease, paint	Foam, CO2, dry chemical (smother)	**B** = **B**oils / **B**arrel
C	Energized electrical equipment, insulation	CO2, dry chemical (non-conducting)	**C** = **C**urrent
D	Combustible metals (magnesium, sodium)	Special dry powder	**D** = **D**ense metal

MEMORY HOOK **Engine room = B and C**

An engine compartment is full of fuel and lube oil (**B**) and live wiring (**C**), so engine room fires are Classes B and C. CO2, dry chemical and foam are all suitable for a Class B engine-space fire ("all of the above").

How fire spreads: heat transfer



CONDUCTION
through metal by contact

CONVECTION
hot gases moving through vents

RADIATION
heat rays across open space

Prevent spread by cooling adjacent surfaces (bulkheads and decks). **Close ventilation** to a burning space to stop convection and cut off oxygen. Radiation is the one method that needs no contact or moving air.

Firefighting II: agents and extinguishers

Agent	Works mainly by	Classes	Watch out for
Water (solid stream)	Cooling; stream breaks up burning solids	A	Never on oil/gasoline: splashes and spreads burning fuel
Water fog / spray	Cooling (most effective cooling agent)	A (also shields crew)	Water absorbs the most heat of any agent
Foam	Smothering; cools as a secondary effect	A, B	Conducts electricity: not for Class C
CO2	Smothering (displaces oxygen)	B, C	Little cooling, so reflash risk; suffocation in small spaces; horn gets extremely cold
Dry chemical (BC)	Breaking the chain reaction	B, C	Fast knock-down, little cooling
Dry chemical (ABC)	Breaking the chain reaction	A, B, C	Advantage over CO2: multipurpose

Portable extinguisher facts the bank loves

- **Sizes I to V:** I is the smallest. Sizes I and II are **portable** (carried by hand). Over **55 lb** charged = **semi-portable** (III to V, often wheeled or on a hose reel).
- **CO2 has no gauge.** Check it by **weighing** on a calibrated scale against the stamped full weight; recharge if more than 10% low. Coast Guard inspectors weigh CO2 units.
- **Cartridge-operated dry chemical:** at annual servicing, **weigh the cartridge**.
- **Partially discharged** = treat as empty: label it and recharge as soon as possible.
- **Name plate** (uninspected vessels) shows item name, rated capacity and manufacturer ID. It does **not** show the hydrostatic test date.
- **Fixed CO2 systems** are not used in crew quarters (lethal) or in spaces open to the atmosphere (gas blows away).

Tactics and FIRST actions

Situation	Keyed answer
You discover a fire	**Sound the alarm** first, every time
Heavy smoke from the paint locker	**Secure the ventilation** first (starve it; don't open the door)
Oil or gasoline fire in the bilge	Do **NOT** use a solid stream of water
Class B fire	Best fought by keeping oxygen from the fuel (smother)
Class C fire	Agent must be **non-conducting**
Fire blanket on a small fire	Removes **oxygen**
Fire hose at hydrant	Stays connected with nozzle, except in heavy weather or when cargo work could damage it
Hose couplings	Bronze, brass or soft alloy (corrosion resistant, non-sparking)

MEMORY HOOK **Spontaneous combustion: oily rags in a pile, fire in a while**

Caused by **chemical action within a substance** (oxidation makes its own heat), classically **rags soaked in linseed oil**, and by careless storage or disposal. **Good housekeeping** prevents fire by eliminating fuel. Charging **batteries** give off hydrogen: prevent buildup with natural or mechanical ventilation.

Six-pack regulations (uninspected passenger vessels)

As an OUPV operator you'll run an **uninspected passenger vessel**: six or fewer passengers for hire. These questions are labeled "(Uninspected Vessel Regulations)" and come from 46 CFR Subchapter C.

SIX-PACK RULE **Who can run the boat**

Each self-propelled UPV carrying not more than six paying passengers must be **under the direction and control of an individual licensed by the USCG**. That's your OUPV. A Merchant Mariner Credential is valid for **5 years**.

Portable fire extinguishers on motorboats (46 CFR 25.30-20)

Length	No fixed system	With fixed system in engine space
Under 26 ft	1 × 5-B	0
26 ft to under 40 ft	2 × 5-B	1 × 5-B
40 ft to 65 ft	3 × 5-B	**2 × 5-B**

One 20-B may replace two 5-B. The number is based on the vessel's **length**. Older questions may use the former B-I / B-II ratings for the same idea.

MEMORY HOOK **40 to 65 = three, minus one with a fixed system**

The bank's question: 45-ft motorboat built in 2017 with a fixed system needs **two 5-B**.

Placards and lists

Required item	Contains	Does NOT contain (the EXCEPT answer)
Safety orientation placard	How to don a PFD, where PFDs are stowed, where the emergency checkoff list is	The vessel's emergency radio call sign
Emergency checkoff list	Specific emergencies: man overboard, fire, heavy weather / hazardous bar crossing	Generic "actions in the event of accident"
Garbage placard (26 ft and over)	Summary of discharge restrictions, posted prominently	-

Lifesaving gear on a six-pack

- Each person aboard needs an approved **wearable life jacket** (Type I). Cushions and vests don't satisfy it.
- All PFDs must be **Coast Guard approved**, and life jackets are **marked with the vessel's name**.
- Life jackets and all lifesaving gear must be **readily accessible**: never locked up, never still in the plastic wrap.
- A life jacket light with a non-replaceable battery is replaced **on or before the battery's expiration date**.
- A documented vessel shows its **name on both bows** and **name plus hailing port on the stern**.

Lifesaving equipment I: PFDs, suits, EPIRBs

PFD types

Type	Name	Use	Key fact
I	Offshore life jacket	Open, rough, remote water	Most buoyancy; turns most unconscious wearers face-up. Required for six-pack passengers
II	Near-shore vest	Calm inland water	Turns some unconscious wearers face-up
III	Flotation aid	Protected inland water, quick rescue expected	Comfortable and mobile, but won't reliably turn an unconscious person
IV	Throwable	Ring buoy, cushion	Thrown, not worn
V	Special use	Work vests, hybrids	Only counts when worn / used as labeled

Immersion suits

- Must turn an unconscious person's mouth clear of the water **within 5 seconds**, with no help.
- Must allow walking, climbing a ladder and picking up small objects before abandoning.
- Equipped with a **whistle, a light and retroreflective material**. Not fireproof; heat damages them.
- Protects against **hypothermia** (distractor: hypoglycemia).
- If you must jump: inflate the external flotation bladder **after** you're in the water.

EPIRB

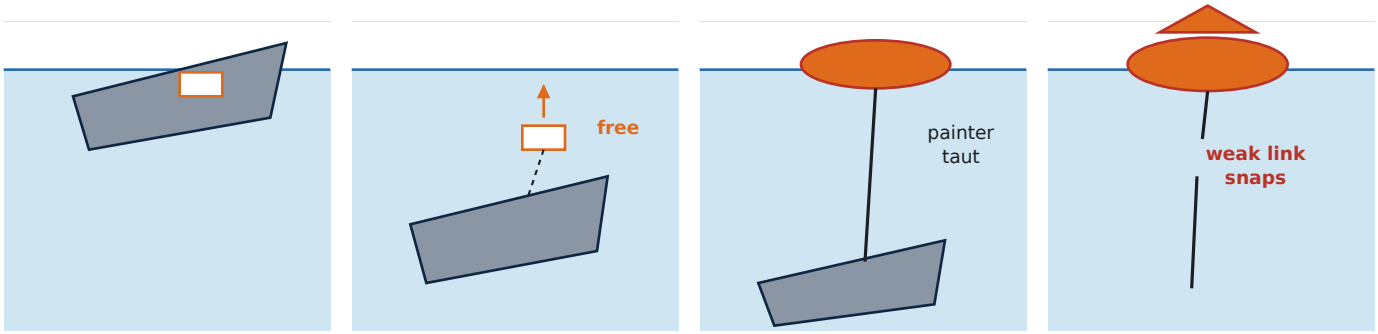
Question	Answer
What it stands for	Emergency Position Indicating Radio Beacon
Who must carry a Category I 406 MHz EPIRB	Vessels on the **high seas** or beyond **3 miles** from the Great Lakes coastline
What Category I means	**Floats free** of a sinking vessel and **activates automatically**
How often to test	**Monthly** (self-test; also check battery and hydrostatic release dates)
When to activate	**MAYDAY-type** emergencies only (grave and imminent danger)
What it sends	Continuous signal with its unique ID, used to identify the vessel and locate the beacon

TIP **Visual distress signal for daylight only**

Hand-held orange smoke is day-only. Red hand flares and parachute flares are day-and-night.

Lifesaving equipment II: inflatable liferafts

Float-free deployment



1 Vessel sinks
raft canister in cradle

2 HRU trips
water pressure at 10–15 ft

3 Painter pulls
CO2 fires, raft inflates

4 Weak link parts
raft floats free

Stow so the raft floats free and inflates automatically if the vessel sinks.

Part	What it does
Hydrostatic release (HRU)	Water pressure at about 10–15 ft frees the canister from its cradle
Painter / operating cord	Secured to the cradle or a strong point through a weak link; pulling it fires the CO2
Weak link	Breaks so a vessel sinking in water over 100 ft can't drag the raft under
Sea anchor (drogue)	**Deploys automatically** on inflation: reduces drift, adds stability
Canopy lights	Come on **automatically** when the raft inflates
Safety knife	In a special pocket **near the forward entrance**, to cut the painter
Canopy	Shade: cuts sweating and sun exposure, so it reduces dehydration

Handling and servicing

- **Launch by hand:** throw the raft over the side, then pull the operating cord sharply.
- **Board dry** if at all possible. Staying out of the water preserves body heat.
- **Steel bands** on the canister: **leave them in place**. They hold the halves together and part on inflation.
- Inspection, testing and painter renewal are done only by a **Coast Guard approved servicing facility** (typically annually). The crew never opens the canister.
- Rafts carry a safety knife, a pump to top up the tubes and an external lifeline: **all of the above**.

EXAM TRAP **Steel bands: two versions of the same question**

One version keys "leave the bands in place." A newer version keys "leave them intact and report it to the servicing authority." Either way, the crew doesn't cut or remove them.

Survival at sea

Question	Keyed answer	Why
FIRST action after abandoning and clearing the vessel	**Identify the person in charge**	Organization drives everything else
Several rafts in the water	**Secure the rafts together**	Bigger target, shared supplies
When to first drink (supplies limited)	**After 24 hours**	Water drunk on day one is lost as urine
Survival without water (conserving moisture)	**8 to 14 days**	Shade, rest, no sweating
Drinking seawater	**Under no conditions**	Accelerates dehydration
Take seasickness pills when?	**As soon as possible** after boarding	Prevent vomiting and fluid loss
Greatest danger in cold water / high latitudes	**Hypothermia**	Water pulls heat ~25× faster than air
Swimming through an oil fire	Do **NOT** swim with the wind	Swim into or across it so flames blow away

MEMORY HOOK **Survival order: people, rafts, then water**

Take charge → tie rafts together → stay dry and warm → no water for 24 hours → ration after that. Hypothermia kills long before thirst or hunger.

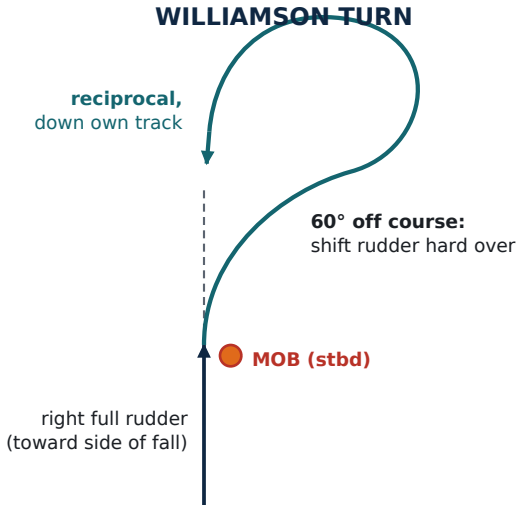
BANK KEY **Fresh water from fish**

Survival manuals do list fish fluids as an emergency source, but the bank marks "fresh water may be contained in some fish" as the **FALSE** statement. Pick it when asked which statement is false.

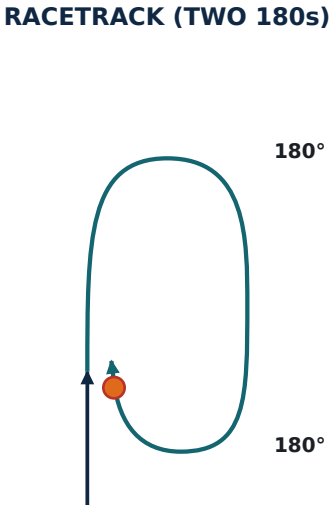
Man overboard

Situation	Keyed answer
Person falls from the **port** side	**Hard left rudder** immediately (swings stern and prop away)
Person falls from the **starboard** side	**Full right rudder**; throw a life preserver at once
Don't know which side	**Stop the propellers** and throw a ring buoy
You see someone go over	**Call for help and keep them in sight** (point continuously)
Key to a successful rescue	**Preparation, practice and alertness**
Bowline in a rescue	Fixed loop that won't constrict; good for hoisting an unconscious person

Man overboard: recovery turns



Person NOT in sight · night · fog



Person IN sight · cold water · fastest

Turn	How	Use when
Williamson	Rudder hard over toward the side of the fall. At **60°** off the original course, shift the rudder hard to the other side and come around to the reciprocal.	Person **not in sight**: night, fog, or the time of the fall is unknown. Puts you back on your own track line, heading the other way.
Racetrack	Two 180° turns with a straight leg between.	Person is **visible** and the water is **very cold**: gets back to them fastest.

MEMORY HOOK **Williamson = Without sight**

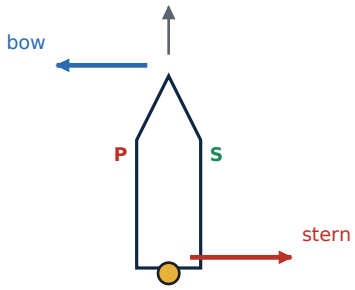
Lost them in the fog or at night? Williamson brings you back down the same path. Can still see them? Racetrack.

BANK KEY **Radio call for an unrecovered person overboard**

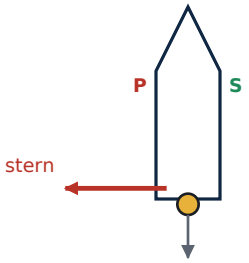
Many authorities would say MAYDAY, but the bank keys **PAN-PAN** (urgency) because the vessel itself isn't in distress.

Boat handling I: props, pivot points and turning

Prop walk with a right-hand propeller



RH prop, going AHEAD
stern → starboard, bow → port

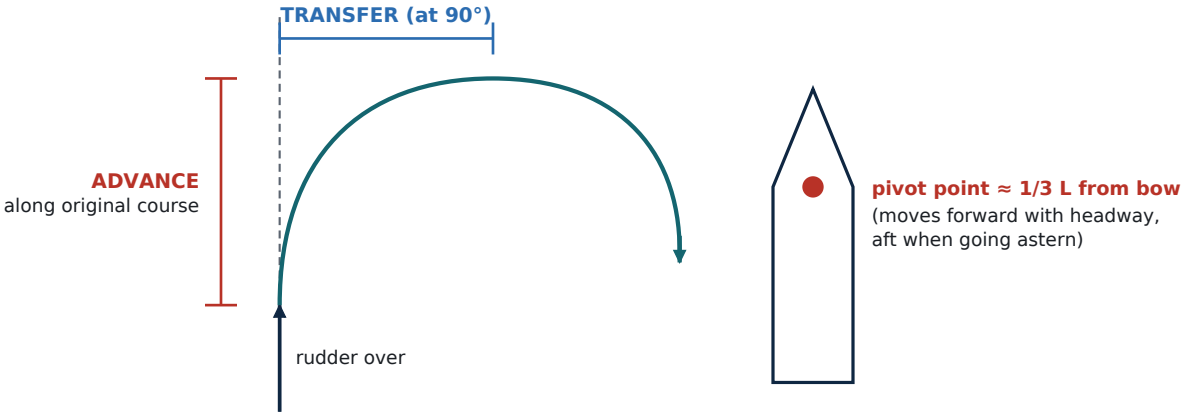


dock on PORT side:
backing pulls stern IN

RH prop, going ASTERN
stern walks → port

- Dead in the water, rudder amidships, RH prop starts **ahead**: bow goes **to port**.
- Landing **port side to** and backing down: stern swings **toward the dock**. That's why port-side landings are easier.
- Twin screw, **port engine lost**: the starboard engine pushes the bow to port, so **compensate with right rudder**.

Turning circle and pivot point



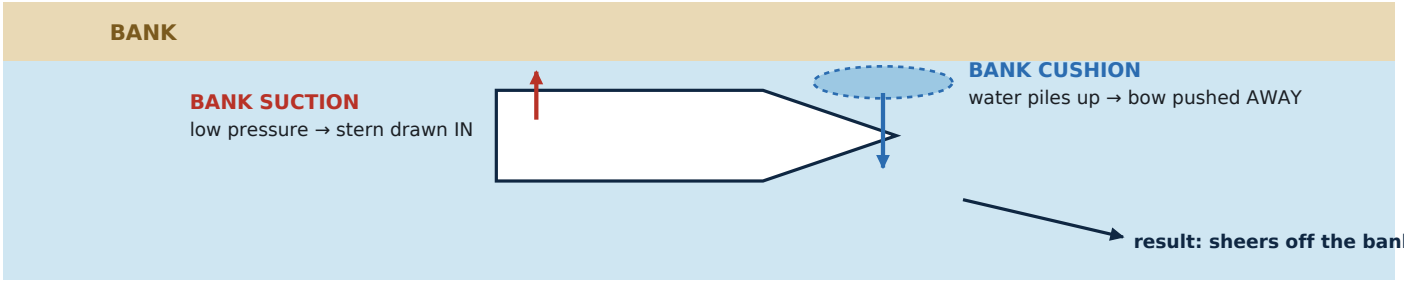
MEMORY HOOK **Advance goes Ahead, Transfer goes Across**

Advance = distance gained in the direction of the original course. **Transfer** = distance gained at right angles to it.

Situation	Keyed answer
Using a buoy four points on the bow to judge a turn; the buoy moves forward	**Decrease the rate of turn**
Turning with a following current	Rudder is less effective: **wider turning radius**
No-wake zone	**Minimum speed to keep steerage**
Hull type with the largest wake at hull speed	**Displacement** hull

Boat handling II: banks, shallows, docking, heavy weather

Bank effect and ship interaction



Two vessels passing close in a narrow channel interact **the same way as bank cushion and suction**: pressure between the bows pushes them apart, low pressure aft draws the sterns together.

Squat

Running into shallow water, **squat** reduces bottom clearance. It increases sharply with speed, and the **stern sucks down lower than the bow**. The fix is to slow down.

Docking and mooring

Question	Keyed answer
Most useful line when maneuvering alongside	**After-bow spring**
Docking: when possible go in	**Against the current**
Most favorable wind and current	**Both parallel to the pier, from ahead**
When is wind effect most noticeable	**When backing**
Leaving a mooring buoy in calm weather	**Back away a few lengths**, then go ahead
Passing a berthed vessel with slack lines	**Reduce speed** first
Reducing wake damage	**Slow down** when passing moored vessels

MEMORY HOOK **Dock into the current like you land into the wind**

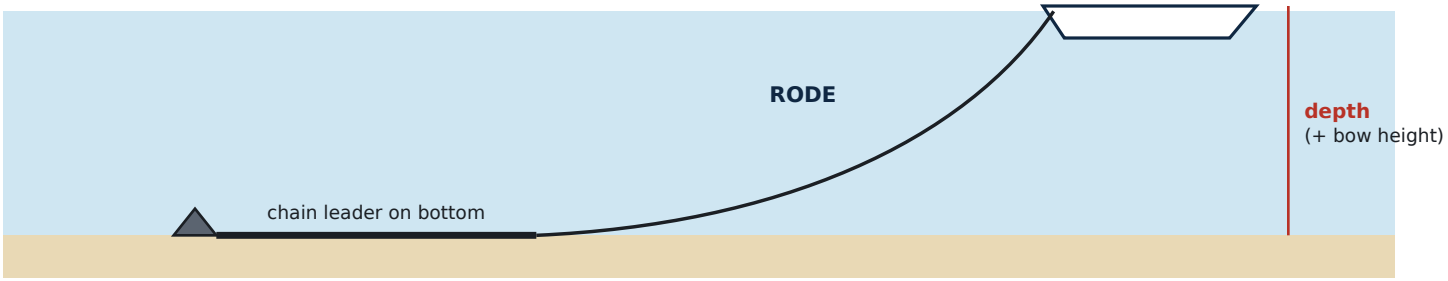
Wind and current from ahead give steerage at low ground speed and act as a brake.

Heavy weather

Situation	Keyed answer
Heave-to in heavy weather	**Take the sea on the bow**, minimum speed to hold position
Broken down and rolling heavily	**Heaving-to** (bank key; really a sea anchor holding the bow to the seas)
Propeller lifting clear and racing	**Decrease speed**
Turned broadside to heavy seas	**Broached** ("broaching to")

Anchoring and towing

Rode = Depth × Scope (normal 5:1 to 7:1)



12 ft × 6 = 72 ft of rode

Question	Keyed answer
Scope is the ratio of	Length of rode to depth of water
Normal length of rode	**5 to 7 times** the depth
Part that digs in and holds	**Fluke** (holding power = the anchor's ability to dig in)
Best small-boat holding bottom	**Mud and clay**
Best time to let go	When making **slight sternway**
Paying out more rode	**Increases holding power** (flatter pull)
Holding increases when	More **chain lies along the bottom**
Protect fiber rode near the anchor	Replace that portion with a **short length of chain**
Best way to detect dragging	Changes in **bearings of fixed objects** ashore
Stop dragging	**Pay out more rode**
Anchor won't break free	**Run in a wide, slow circle** to change the angle of pull
Anchor watch position checks	Every **15 minutes**; purpose: chafe, dragging, traffic (all of the above)

Towing

Term / question	Keyed answer
Catenary	The **downward curve (sag)** of the towline
More catenary	**Reduces shock stress** on the hawser
Towing in rough seas	**Slow and steady with long scope**
Where to secure a towline	A **tow-rated strong point** (bollard or bitt), not a light cleat
Legs on a bridle	**Two**; lengthening them **reduces yawing** of the tow
Shortening the hawser	**Decreases the tug's maneuverability**

Marlinespike seamanship

Knot / splice	Job	Hook
Bowline	Temporary fixed eye; improvised mooring eye; parted mooring line; MOB loop; hoisting an unconscious person	King of knots: won't slip or jam
Square knot	Join two lines of **equal** size	Equal = square
Sheet bend / becket bend	Join lines of **different** size (small line to a larger one)	Different = sheet
Sheepshank	**Shorten** a line	Sheep get sheared short
Figure eight / overhand	**Stopper**: keeps the end from unreeving through a block; temporary anti-unravel	Bulky end
Cleat hitch	Round turn, then figure eights, then a locking turn	Secure yet quick release
Eye splice	**Permanent loop**	Eye = loop
Short splice	**Permanently join** two lines; strongest join; weakens line the LEAST	Short joins two short pieces
Long splice	Join lines that must run through a block or sheave	Short splice is too fat for a sheave
Whipping	Twine around the end to keep it from unraveling / fraying	-

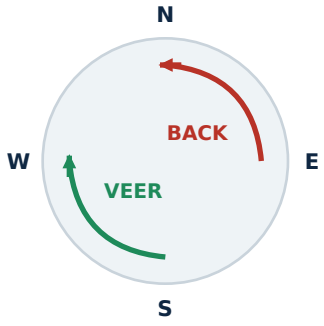
Line types

Line	Key fact
Nylon	Stretches the most (great shock absorber); **dangerous** because it stretches and snaps back
Polypropylene	**Floats**: rescue and ring-buoy lines
All synthetics	Strength and durability most impaired by **sunlight** (UV)

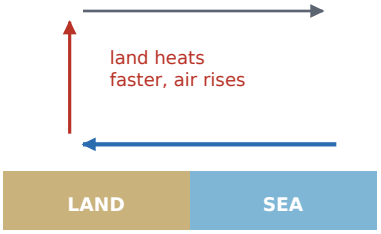
Deck terms

Term	Meaning
Small stuff	Line under **1.5 in** circumference (bank key)
Chafing gear	Wrapping at all wearing points of mooring lines to prevent abrasion
Bollard	Stump-like metal post on a pier for mooring lines
Dipping the eye	Pass the second eye up through the first so either line lifts off a bollard first
Crossing the springs	Longer springs leave the boat free to rise and fall with the tide
Fid	Tapered wooden pin for splicing heavy fiber rope (marlinespike is steel, for wire)
Gantline	Rope rove through a tail block to hoist a bosun's chair
Safety shackle	Identified by the **design of the pin** (bolt, nut and cotter)

Marine weather

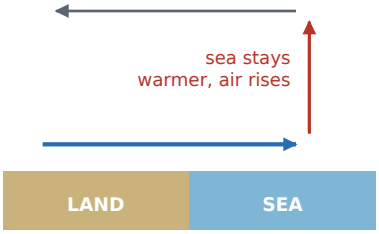


Veer = clockwise (S→W) · Back = counter (W→S)



SEA BREEZE · day

named for where it comes FROM



LAND BREEZE · night

named for where it comes FROM

Topic	Keyed answer
Boundary between two air masses	**Front**
Cold air displacing warm air	**Cold front**: brief showers, strong shifting winds along or ahead of it
After a cold front passes	Pressure **rises**, often quickly, with clearing skies
Low pressure approaching	Weather generally **worsens** (falling barometer)
Generally circular low	**Cyclone** (a high is an anticyclone)
Thunderstorm cloud	**Cumulonimbus**
Dew point	Temperature at which the air becomes **saturated**
Warm moist air over cold water or land	**Advection fog** (radiation fog: land, clear calm nights)
Estimate wind speed from sea state	**Beaufort** scale
Wind felt on a moving vessel	**Apparent wind**; zero when the true wind is from astern at your speed
U.S. unit for pressure	**Inches** of mercury (standard 29.92)
Greatest challenge to seamanship	Waves, seas and surf: **all of these**

MEMORY HOOK **Veer = clockwise, Back = counterclockwise**

Think of a clock hand veering forward. A **backing** wind goes backward (counterclockwise). Both definitions are for the Northern Hemisphere.

Marine radio

Priority signals

MAYDAY ×3	**DISTRESS**	grave & imminent danger to vessel or life; immediate help needed
PAN-PAN ×3	**URGENCY**	urgent concern for safety of a person or vessel, not imminent
SÉCURITÉ ×3	**SAFETY**	navigation/weather info: aid off station, storm warning, hazard

EXAM TRAP **Which word for which message**

An aid to navigation off station = **SÉCURITÉ**. A Coast Guard call to all stations starting "Pan-Pan" = **urgency signal**. Urgent concern for the safety of a person = **PAN-PAN**.

Channels and rules

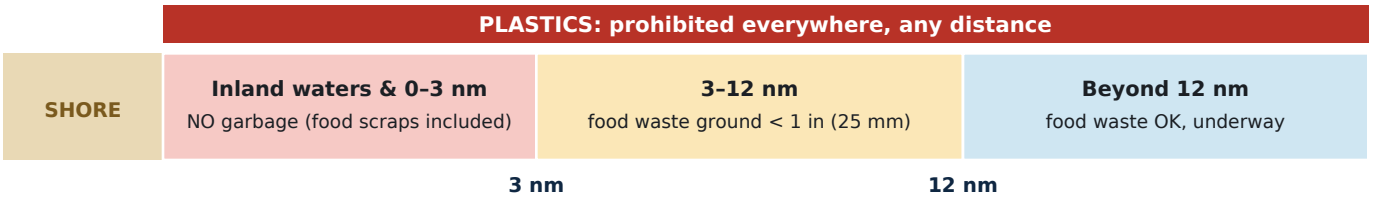
Question	Keyed answer
Distress, safety and calling	**Channel 16** (156.8 MHz). Never use it as a working channel
Navigational (bridge-to-bridge) only	**Channel 13** (156.65 MHz)
Bridge-to-bridge rules apply to	Power vessels **20 m** and over; passenger vessels **100 GT** and over carrying passengers for hire; towing vessels 26 ft and over
Max VHF power	**25 watts** (1-watt low setting required)
Typical vessel-to-vessel range	**15 to 20 miles**
Why antenna height beats power	VHF is **line of sight**
No reply; wait before calling again	**Two minutes**
Simplex mode means	**Only one person talks at a time**
Squelch setting	Open until static is heard, then back off just until it stops
Before transmitting on a busy channel	**Listen first**; don't step on traffic
Other station monitors the working channel too	Call on the **working frequency** (keep 16 clear)
Most affected by atmospheric disturbance	**MF/HF** radiotelephone

Pollution prevention

Oil spills

Question	Keyed answer
Who you report any spill or sighting to	**National Response Center** (USCG), 1-800-424-8802, regardless of local rules
Who must report	The **person in charge**, immediately
Penalty for failing to report	Fine and up to **5 years** in prison
After reporting	**Contain and remove** as much oil as possible
Straw is an example of a	**Sorbent**
Skimmers are	**Mechanical** recovery
Method needing prior approval	**Chemical agents** (dispersants, sinking agents)
Most minor spills are caused by	**Human error**
Small spills	Serious, because the effect is **cumulative**

Garbage (MARPOL Annex V)



MARPOL Annex V simplified for the exam (U.S. inland waters & Great Lakes: no garbage discharge)

- Vessels 26 ft and over post a **garbage placard**; that's how the person in charge keeps the crew informed.
- A required **waste management plan** must be written and describe procedures for **collecting and discharging garbage**.

BANK KEY **What does MARPOL regulate?**

MARPOL covers oil, noxious liquids, sewage, garbage and air, but the bank keys **garbage**.

Sewage and marine sanitation devices

MSD	What it does
Type I	Treats sewage (maceration and disinfection) before discharge
Type II	Treats sewage to a higher standard before discharge
Type III	**Holding tank**: stores sewage at ambient pressure and temperature

Untreated holding-tank sewage may be discharged **beyond 12 nm, underway at 4 knots or more** under MARPOL. U.S. domestic rules use 3 nm, so read the question's wording.

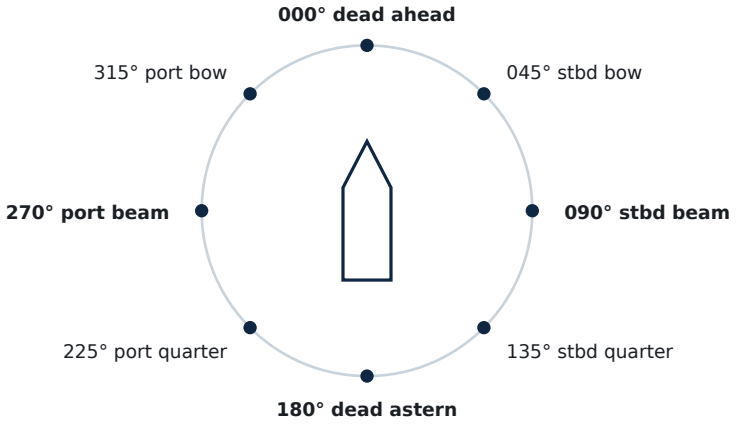
Engines, fueling and watchkeeping

Gasoline safety

- Gasoline vapor is **heavier than air** and settles in the **lowest part of the bilge**.
- Before starting, run the exhaust blower long enough for **at least one complete change of air** (rule of thumb: 4 minutes).
- While fueling, keep the nozzle **in contact with the fill opening** to guard against a static spark.
- Fill tanks **nearly full, leaving room for expansion**.
- After fueling, **open all hatches and ventilate** every compartment.
- The tank shut-off valve, operable from outside the tank space, **shuts off the gasoline supply at the tank**.

Engine question	Keyed answer
Gasoline engine turns over freely but won't start	**Defective ignition system**
Troubleshooting a no-start	Check battery, spark plugs, carburetor, fuel line (no open sparks)
Carburetor's job	**Mix fuel and air**
Diesel ignition	**Heat of compression**, from a high **compression ratio**
Healthy diesel exhaust	**Clear** (black = too much fuel / too little air, blue = oil, white = water)

Lookout and watchkeeping



Measured clockwise from the bow, **not from north.**

Lookout report = object + relative bearing + range/detail, *e.g.* "*Vessel, broad on the starboard bow, about 2 miles, crossing left.*"

Question	Keyed answer
When a lookout may leave the station	Only when **properly relieved**
Statement that transfers the watch	Oncoming officer says **"I relieve you,"** with course per gyro and standard
How to scan	**Slowly and regularly**: left-right-left and top-bottom-top in overlapping steps
Prevent eye fatigue	Periodically focus on a **close object**
Binoculars	Increase distance vision but **narrow the field**; no help in fog
Bright room to dark	Vision **decreases** at first; gradual improvement is **dark adaptation**
Helm watch	Hold the course **within 5°**; ask if an order is unclear
Choosing lookout positions	Safety and ability to do the job; a stable spot so they can't be swept overboard
Assigning crew duties	Match **experience level and abilities**

Your trouble spots

The 17 Deck General questions you missed on MLS

Topic	The fact to lock in
Pollution	MARPOL is keyed as regulating **garbage**
Pollution	Any oil spill or sighting goes to the **USCG National Response Center**
Pollution	Straw used in cleanup is a **sorbent**
Fueling	The remote tank valve **shuts off gasoline at the tank** (it's on/off, not a throttle)
Immersion suit	Turns an unconscious person face-up **within 5 seconds**
Liferaft	Steel bands on the canister: **leave them in place**
EPIRB	Category I = **floats free and activates automatically**
Liferaft	Launch by hand: **throw it over, then pull the operating cord**
Anchor watch	Confirm position every **15 minutes**
Marlinespike	The **short splice** weakens a line the least
Twin screw	Port engine out, running on starboard: **right rudder**
Survival	First action after clearing the vessel: **identify the person in charge**
Survival	"Fresh water may be contained in some fish" is keyed **false**
MOB	Falls off the port side: **hard left rudder**
MOB	Unrecovered person, need help from other boats: **PAN-PAN**
MOB	Falls off the starboard side: **throw a life preserver** immediately
Six-pack rules	Emergency checkoff list does NOT cover generic **"accident"** actions

Bank keys: memorize the expected answer

Question	Bank answer	Why it's debatable
Line whose primary job is to reduce fore-and-aft motion (Q170)	**Bow and stern lines**	Springs are usually taught as the surge lines
Small stuff cutoff	**1.5 in** circumference	Some references say 1-3/4 in
Helm course tolerance	**Within 5°**	Another item treats 10° as acceptable
Broken down, rolling in heavy seas	**Heaving-to**	Without power you'd use a sea anchor
Unrecovered person overboard	**PAN-PAN**	Many would call MAYDAY
Fresh water in fish	**False** statement	Manuals list fish fluids as a source
What MARPOL regulates	**Garbage**	It covers oil, sewage, air and more too
Liferaft steel bands still on	**Leave intact and report**	Bands left on can stop float-free

Self-test (answers on next page)

1. You find a fire in a pile of linen. What class is it, and what is the best agent?

2. Name the four elements of the fire tetrahedron. Which one does dry chemical attack?

3. Why shouldn't you use a solid stream of water on a gasoline fire in the bilge?

4. What's the main hazard of discharging CO2 in a small enclosed space?

5. How do you check that a CO2 extinguisher is fully charged?

6. Heavy smoke is coming from the paint locker. What's your FIRST action?

7. How many 5-B extinguishers does a 30-ft motorboat with a fixed engine-space system need?

8. What's missing from the safety orientation placard on a six-pack?

9. Which PFD type is required for each person on a six-pack carrying passengers?

10. How often must the EPIRB be tested, and when should it be activated?

11. What protects a liferaft from being dragged under by a vessel sinking in deep water?

12. When should you first drink water from limited supplies in a liferaft?

13. At night a crewmember may have gone overboard some time ago. Which turn do you use, and why?

14. A crewmember falls off the port side. What do you do with the rudder?

15. Dead in the water, RH prop, rudder amidships. You go ahead. Which way does the bow go?

16. In a narrow channel, which way do bank cushion and bank suction push the bow and the stern?

17. What happens to squat as speed increases in shallow water?

18. How much rode do you need for a 6:1 scope in 15 ft of water (ignore bow height)?

19. Your anchor is dragging. What's the best first fix?

20. Which knot joins two lines of different sizes? Which shortens a line?

21. Which line floats? Which stretches the most?

22. A forecast says the wind will back. Which way will it shift?

23. What's the difference between Channel 16 and Channel 13?

24. You spill oil. Who do you notify, and what's the maximum prison term for not reporting?

25. A gasoline engine cranks freely but won't start. What's the most likely cause?

Answer key

1. Class **A** (leaves ash). **Water**.

2. Heat, fuel, oxygen, **chain reaction**. Dry chemical breaks the chain reaction.

3. It **splashes and spreads** the burning fuel, which floats on the water.

4. Suffocation: it displaces oxygen for people as well as the fire.

5. Weigh it and compare with the stamped full weight (no gauge).

6. Secure the ventilation.

7. One (26 to under 40 ft = two, minus one for the fixed system).

8. The vessel's **emergency radio call sign**.

9. Type I wearable life jacket.

10. Tested **monthly**; activated only for **MAYDAY-type** emergencies.

11. The **weak link** in the painter.

12. After 24 hours (except for the injured).

13. Williamson: it returns you on a reciprocal course down your own track line.

14. Hard left rudder (toward the side of the fall, swinging the stern away).

15. To port (stern walks to starboard).

16. Cushion pushes the **bow away** from the bank; suction pulls the **stern toward** it.

17. Squat **increases** sharply, and the stern sucks down lower than the bow.

18. 90 ft (15 × 6).

19. Pay out more rode.

20. Sheet (becket) bend. **Sheepshank**.

21. Polypropylene floats. **Nylon** stretches the most.

22. Counterclockwise (backing).

23. 16 is distress, safety and calling; 13 (156.65 MHz) is bridge-to-bridge navigation only.

24. The **National Response Center** (USCG); up to **5 years**.

25. A **defective ignition system**.

> **TIP** **Scoring yourself**
>
> The module needs 70%, but aim for **23/25** here. Any miss: reread that page, then run that topic in the app's Quiz mode until it's clean.